



Abstract Book

The First World Conference
Fighting COVID-19 Pandemic
Health Challenges

**COLLECTIVE KNOWLEDGE.
GLOBAL HEALTH.**

26-28 March 2022, Belgrade, Serbia



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Comparison of Three Commercial Kits for SARS-CoV-2 RNA Detection

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Background: The detection of Severe Acute Respiratory Syndrome Coronavirus-2 (SARS-CoV-2) early, quickly, and reliably is critical for the successful control of coronavirus disease 2019 (COVID-19). The quantitative real-time reverse transcription-polymerase chain reaction (RT-PCR) assay is considered the gold standard for molecular diagnosis of SARS-CoV-2. The objective of this study was to compare the clinical performances of the three authorized tests – the Abbott Real Time SARS-CoV-2 (ACOV) assay (Abbott Molecular Inc., North Chicago, IL), the GeneFinder™ COVID-19 Plus RealAmp (GeneFinder) Kit (OSANG Healthcare Co., Ltd, Dongan-gu Anyang, Korea) and the Biomerieux ARGENE® SARS-CoV-2 R-GENE® real-time detection (ARGENE) kit (bioMérieux SA., Marcy l'Étoile, France) and to determine whether the selection of targeted genes has an impact on the test's specificity.

Methods: We included 233 nasopharyngeal swabs (NPS) from adult individuals with symptoms or suspected of COVID-19, aged from 18 to 93 years, previously tested by the ACOV and subsequently tested by the GeneFinder and the ARGENE.

Results: We found that the GeneFinder assay detected the most cases of COVID-19 infection, followed by the ACOV assay, and then by ARGENE. Positive agreement ranged from 73.46% to 95.29%, with the strongest level of agreement observed between the GeneFinder and ACOV assays and the lowest agreement between the GeneFinder and ARGENE assay. The negative percent agreement was 100% (GeneFinder/ACOV, GeneFinder/ARGENE, and ACOV/ARGENE), with 3.1% of cases being false-negative using the ACOV test and 17.9% of samples being false-negative using the ARGENE assay to detect SARS-CoV-2.

Conclusions: Due to possible false-negative results using the ACOV and ARGENE tests, we recommend complete testing with the GeneFinder test.

Keywords: COVID-19, SARS-CoV-2; RT-PCR, RNA isolation; Virus detection

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